

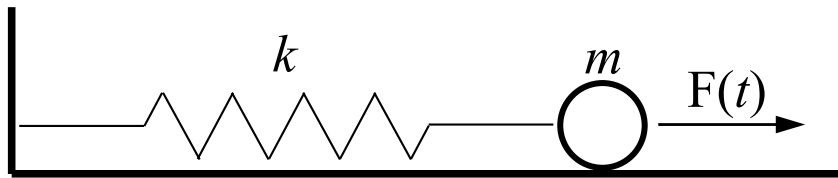
1. Consider two independent vectors $\mathbf{A}$ and $\mathbf{B}$.

$$\mathbf{A} = (A_x, A_y, A_z), \quad \mathbf{B} = (B_x, B_y, B_z),$$

- A) Rewrite vector $\mathbf{A}$ with unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. Vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are unit vectors which are parallel to x , y and z directions in the Cartesian coordinates, respectively.
- B) Write scalar products of $\mathbf{A}$ and $\mathbf{B}$ with A_x, A_y, A_z, B_x, B_y and B_z .
- C) Write vector products $\mathbf{A} \times \mathbf{B}$ with A_x, A_y, A_z, B_x, B_y and B_z . You can use unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

2. A spring with a natural length of L and spring constant of k is connected to a wall and a mass point (mass m). Force $F(t)$ is applied to a mass point.

- A) Write the equation of motion for the mass point.
- B) Find the motion of the particle as a function of t , when $F(t) = 0$ and the initial condition is $x = x_0, v = 0$ (at $t = 0$).
- C) Find the motion of the particle as a function of t , when $F(t) = F_0 \sin \omega_0 t$ and the initial condition is $x = 0, v = v_0$ (at $t = 0$). F_0 and ω_0 are constants.

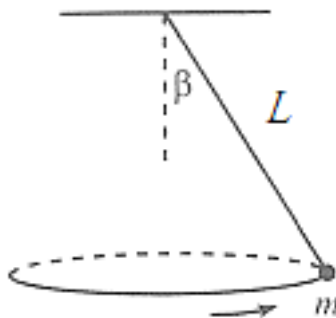


3. In Cartesian coordinates, a general vector takes the following form. $\mathbf{P}$ is position vector, $\mathbf{i}$ and $\mathbf{j}$ are unit vectors which are parallel to x and y direction, respectively. r and θ change with time t . $\mathbf{i}$ and $\mathbf{j}$ does not change with time t .

$$\mathbf{P} = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j}$$

- A) Write the time derivative of $r \cos \theta$ with respect to time t . You can use the Newton's notation in your solution.
- B) Write the time derivative of $\mathbf{P}$ with respect to time t . You can use the Newton's notation in your solution.

4. A mass hangs from a massless string of length L . Conditions have been set up so that the mass swings around in a horizontal circle, with the string making a constant angle β with the vertical. What is the angular frequency ω of this motion.



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